

Beyond Representation: Index Structure Limitations in Dense Retrieval

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Why this talk

- ▶ Dense retrieval is everywhere: e-commerce, enterprise search, RAG.
- ▶ Practical deployments rely on **approximate nearest-neighbor (ANN)** indices (HNSW, IVF) for speed.
- ▶ Most research focuses on the *representation*—can the embedding capture the right semantics?
- ▶ This talk: **the index itself imposes a separate, quantifiable failure mode** that disproportionately attacks a specific population of queries.

Headline finding

Even when the embedding has enough representational capacity, switching from exact (Flat) to HNSW search forfeits $\sim 63\%$ **of achievable Recall@10** on identifier queries—vs. less than 2% on standard queries.

The two camps

Sparse retrieval (BM25)

- ▶ Inverted index over tokens.
- ▶ Exact lexical matching.
- ▶ Strong on rare terms, IDs, named entities.

Dense retrieval

- ▶ Continuous embedding + nearest-neighbor search.
- ▶ Strong semantic generalization.
- ▶ Indispensable in production for fluency & multilinguality.

- ▶ Prior theory (Tay et al. 2020; Fan et al. 2022): *representation capacity* of dense models can match sparse—given enough dimensions.
- ▶ But: practical dense retrieval uses ANN, not exact search.

What does that cost?

“Spear-fishing” queries

Definition (operational)

Queries whose relevance hinges on a single, low-frequency, high-IDF token: identifiers (SKUs, part numbers), rare terms, precise named entities.

- ▶ In production: a small fraction of total traffic.
- ▶ But operationally critical: failures yield empty search-result pages, abandoned carts, broken RAG retrievals.
- ▶ In our experiments:
 - ▶ **ID subset**: doc identifier used as query (extreme case).
 - ▶ **Rare subset**: uni/bigram terms appearing in ≤ 5 docs.
 - ▶ **Standard subset**: real user queries (control).

The mechanism: centroid dilution (theory)

IVF clusters the corpus into N cells; at query time only the closest n_{probe} centroids are searched.

Centroid of cluster C : $c = \frac{1}{|C|} \sum_{d \in C} f(d) = \dots + \alpha_{t^*} p \phi(t^*) + \dots$

For a query dominated by rare token t^* :

$$\langle c, q \rangle \approx \alpha_{t^*}^2 p = \log^2\left(\frac{M}{n_{t^*}}\right) \cdot \frac{n_{t^*, C}}{M/N}$$

- ▶ α_{t^*} is large for rare tokens (high IDF).
- ▶ But p (in-cluster prevalence) is small for rare tokens.
- ▶ JL projection adds noise floor $\epsilon \approx \sqrt{\log V/d}$.
- ▶ Rare-token centroid signal lost when $\alpha_{t^*}^2 p \lesssim \epsilon$.

Larger embedding d helps ϵ , but cannot fix small p .

Experimental setup

Datasets

- ▶ **Home Depot:** 54,667 product titles
- ▶ **MS MARCO Passage v1.1:** 82,360 passages

Three query subsets per dataset

(1,000 queries each)

- ▶ Standard | Rare (≤ 5 docs) | ID

Representation

- ▶ Char-trigram + JL projection
- ▶ $d \in \{256, 512, 1024\}$
- ▶ Unsupervised

Indices (FAISS)

- ▶ Flat (exact)
- ▶ IVF (nlist, nprobe sweep)
- ▶ HNSW ($M=64$, efC/efS)
- ▶ BM25 (Pyserini)

Why unsupervised trigram, not DPR / Sentence-BERT?

A supervised encoder abstracts away the lexical detail that *defines* a spear-fishing query. The Flat baseline would already fail on IDs—conflating representation failure with indexing failure. We need a representation that *provably* preserves lexical distances (Johnson–Lindenstrauss).

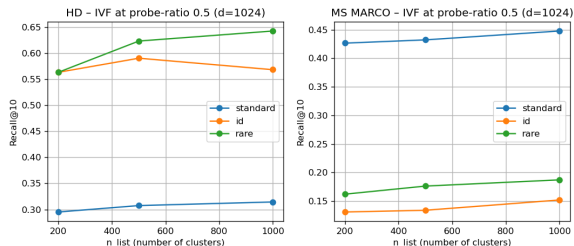
Headline result: the gap

Dataset	Subset	Flat	IVF	HNSW	BM25
HD	standard	0.327	0.314	0.321	0.447
HD	rare	0.751	0.642	0.619	0.878
HD	id	0.874	0.590	0.327	1.000
MS MARCO	standard	0.471	0.448	0.446	0.770
MS MARCO	rare	0.205	0.187	0.166	0.688
MS MARCO	id	0.174	0.152	0.071	—

Recall@10 at $d=1024$, across-token trigrams, best ANN sweep config.

- ▶ Flat→HNSW gap (relative): standard $\approx$ **2%** rare $\approx$ **18%** **id $\approx$ 63%**.
- ▶ Same monotone pattern on both datasets.
- ▶ All gaps statistically significant (paired bootstrap 95% CI excludes zero).

Indirect evidence: IVF cluster granularity



Theory predicts: smaller clusters $\Rightarrow$ less centroid dilution $\Rightarrow$ rare/ID queries benefit more from finer clustering.

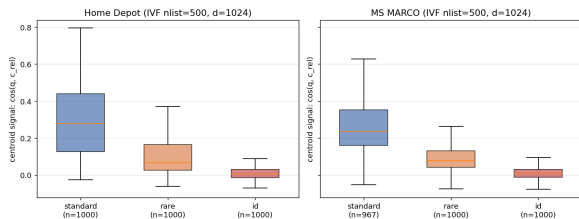
Data: rare & ID gain $2-3\times$ more (relative) than standard going from $n_{\text{list}}=200 \rightarrow 1000$.

Asymmetric improvement is exactly what Eq. (8) predicts.

Direct evidence: measure the centroid signal

For each query, compute the centroid signal $\langle q, c_{\text{rel}} \rangle$ on the centroid of the query's relevant cluster (IVF, $n_{\text{list}}=500$, $d=1024$).

Centroid signal for relevant cluster, by query subset



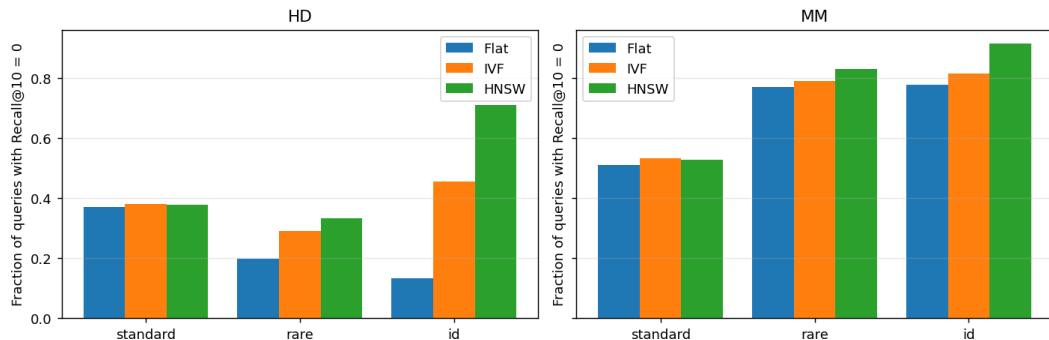
Subset (HD)	signal	rank	hit@250
standard	0.279	2	97%
rare	0.066	69	81%
id	0.007	216	59%

$\sim 40\times$ collapse from standard to ID:
centroid signal disappears into the JL noise floor exactly as Eq. (8) predicts.

For HD-ID queries, 41% of relevant clusters fall outside the top-half by centroid score—*structurally unrecoverable*.

The empty-result rate

Empty-result rate: fraction of queries that retrieve ZERO relevant docs in top-10



- ▶ HD-id Flat: **13%** of queries return nothing relevant in top-10.
- ▶ HD-id **HNSW: 71%**.
- ▶ Standard subset: barely moves.
- ▶ *This is the production-visible symptom.*

“But this is only because the embeddings are simple”

Reviewer question we anticipated: *do learned dense encoders avoid this?*

Re-ran the full grid with all-MiniLM-L6-v2 (Sentence-BERT, 384-dim, supervised).

Dataset	Subset	Trigram Flat R@10	MiniLM Flat R@10
HD	standard	0.327	0.298
HD	rare	0.751	0.428
HD	id	0.874	0.010
MS MARCO	standard	0.471	0.894
MS MARCO	rare	0.205	0.226
MS MARCO	id	0.174	0.001

- ▶ MiniLM nearly doubles MS MARCO standard recall—as designed.
- ▶ MiniLM **cannot distinguish identifiers**: “12345” $\approx$ “67890”.
- ▶ Representation collapses *before* the index has a chance to bias.
- ▶ This is exactly why a supervised encoder is the *wrong instrument* for our study—and exactly why hybrid retrieval is needed in production.

What this means for production

Translation: lost user sessions

Switch from exact to HNSW at fixed embedding quality, you forfeit:

- ▶ $\sim 63\%$ of achievable recall on identifier queries,
- ▶ $\sim 18\%$ on rare-term queries,
- ▶ $\sim 2\%$ on standard queries.

Average-case recall reported with an ANN deployment substantially *overstates* worst-case service quality.

Recommendations:

1. *Query-type-aware routing*: send spear-fishing traffic to BM25 / Flat.
2. *Index-aware training*: penalize placement of rare features in poorly-reachable embedding regions.
3. *Hybrid retrieval*: sparse for precision on rare tokens, dense for semantic generalization. Not new advice—but now with a quantitative justification.

Takeaways

1. Representation capacity is *necessary but not sufficient* for production dense retrieval.
2. ANN indices introduce a *second*, structurally different failure mode that concentrates on rare-token / identifier queries.
3. We formalized this as **indexing bias** and derived a **centroid-dilution bound** that we then directly measured per query.
4. The Flat→HNSW gap on identifier queries is $\sim 30\times$ larger than on standard queries, in both retail and open-domain text.
5. Supervised encoders don't escape this—they collapse at the representation stage on the same queries.

Code & experiments: github.com/alemagnani/sparse-dense-paper

Thank you

Questions?

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